

# Ben Bubnick

## Lead Analytics Engineer - Healthcare Data Modeling and Quality

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### Professional Summary

Healthcare analytics engineer with 10+ years transforming complex clinical, claims, eligibility, and revenue-cycle data into trusted production datasets. Advanced hands-on experience with SQL, dbt, Spark, Python, AWS Athena and Redshift, Snowflake, data-quality test design, root-cause analysis, and client-facing validation. Builds reusable models and controls across ingest, staging, and analytics layers; translates ambiguous findings into clear technical and stakeholder action in HIPAA-regulated environments.

### Technical Skills

**Analytics Engineering:** Advanced SQL, complex CTEs, window functions, aggregations, performance tuning, dbt models/tests/macros/YAML documentation, incremental ELT, SQLMesh, dimensional modeling

**Data Quality:** Null, duplicate, volume, distribution, date, referential-integrity, business-rule, grain, reconciliation, and schema-drift checks; alert triage, root-cause analysis, QA runbooks

**Healthcare Data:** Clinical and EHR entities, professional and institutional claims, EDI/837, payer and eligibility, revenue cycle, prior authorization, HEDIS, HL7 v2/v3, FHIR, HIPAA/HITRUST

**Platforms and Delivery:** Spark/PySpark, AWS S3/Athena/Redshift/Glue, Snowflake, Hadoop/HDFS, Hive, Impala, Python, pandas, Jupyter, Git/GitHub, CI/CD, Jira, Docker, Airflow concepts

### Professional Experience

#### HHAAeXchange

Oct. 2025 - Jun. 2026

##### Healthcare Data Integration Consultant - Contract via Veersa Technologies

*Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox*

Led healthcare source analysis, data-quality investigation, and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics dbt models spanning 55 tables across two organizations and two source-system configurations, reducing client onboarding time by 67%.
- Implemented reusable SQL/Jinja transformation logic for lifecycle states, null handling, sentinel dates, and source variations across organization, office, caregiver, and patient domains.
- Created or materially redesigned 30 automated SQL/dbt tests and macros covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.
- Translated ambiguous requirements and data defects into documented mappings, acceptance criteria, onboarding standards, and 12 investigation runbooks for repeatable triage and client delivery.

#### Adonis

Mar. 2025 - Jul. 2025

##### Healthcare Revenue Cycle Data Consultant - Contract via Health Data Movers

*Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker*

Translated payer and revenue-cycle source data into standardized models for repeatable client onboarding, analytics, and reporting.

- Built, documented, validated, and maintained 54 models supporting claims and revenue-cycle analytics for a provider serving 4,598 families across 113 locations and processing 86,388 annual authorizations.
- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
- Traced duplicate-key failures to undocumented source versioning and implemented composite business grains with SHA-256 surrogate keys across 10 charge and payment models.
- Authored an implementation playbook, technical handoff, and QA runbook that standardized promotion and healthcare data-validation workflows.

#### Arkos Health

Jun. 2023 - Jan. 2025

##### Senior Data Analyst

*Technologies: AWS S3, Athena, Redshift, Glue, dbt, PySpark, Docker, Jupyter, Power BI, Tableau*

- Architected a greenfield AWS data lake integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets.
- Instrumented scheduled pipelines with monitoring, alerting, and 10+ failure classifications; authored three AWS/dbt incident runbooks enabling detection within 5-15 minutes and typical recovery within 2-4 hours.
- Integrated Jupyter analytics and authored platform, architecture, and operating documentation, enabling six analysts to use the governed data lake independently for recurring, executive, and BI reporting.

## Merative

Jul. 2022 - Mar. 2023

### Senior Data Scientist

*Technologies: Azure Data Factory, Blob Storage, Functions, SQL Database, Synapse Analytics, Python, SQL, Azure DevOps*

- Engineered a seven-layer Python and SQL migration-acceptance suite covering partition reconciliation, schema checks, nulls, duplicates, field comparisons, aggregates, and business rules, achieving exact HDFS-to-Azure fidelity.
- Led HIPAA-compliant de-identification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.

## IBM

Jun. 2015 - Jul. 2022

Advanced through five roles delivering healthcare analytics, data architecture, client integration, data quality, and large-scale migration validation.

*Technologies: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, PostgreSQL, Docker, Jenkins, Tableau, HL7*

- **Senior Data Scientist, Nov. 2021 - Jul. 2022:** Engineered a reusable Spark, Hive, and Impala analytics framework powering 50+ prior-authorization reports with daily operational execution and periodic regulatory and financial snapshots.
- **Senior Data Architect, Dec. 2020 - Nov. 2021:** Architected Spark and Hadoop pipelines integrating more than 100 billion clinical records. Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations.
- **Data Architect, Nov. 2017 - Dec. 2020:** Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing validation rework by 25% and mapping rework by 45%. Led delivery across 70+ EHR sources.
- **Software Developer and Data Scientist, Nov. 2015 - Nov. 2017:** Engineered an HL7 v2/v3 parsing pipeline that eliminated \$30,000 in annual vendor costs. Directed 36 client integrations while establishing code-review, QA, and issue-resolution standards.

## Education and Credentials

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**Master of Science, Astrophysics,** University of Cincinnati, 2010

**Certificate of Specialization, Data Science,** Johns Hopkins University, 2015 | **IBM Data Science Professional Certificate,** IBM, 2022